



37.1

AVL Tree (1)



37.1.1 AVL Tree: NIELIT 2017 July Scientist B (CS) - Section B: 7

A balance factor in AVL tree is used to check

- | | |
|------------------------------------|---|
| A. what rotation to make | B. if all child nodes are at same level |
| C. when the last rotation occurred | D. if the tree is unbalanced |

nielit2017july-scientistb-cs data-structures avl-tree easy

Answer key

37.2

Algorithm Design (1)



37.2.1 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 37

Linked lists are not suitable data structures for which one of the following problems?

- | | |
|-------------------|----------------------------|
| A. Insertion sort | B. Binary search |
| C. Radix sort | D. Polynomial manipulation |

nielit-2021-it-dec-scientistb data-structures linked-list algorithm-design

Answer key

37.3

Array (4)



37.3.1 Array: NIELIT 2016 MAR Scientist C - Section C: 55

To sort many large objects or structures, it would be most efficient to place

- | | |
|---|--|
| A. them in an array and sort the array | B. pointers to them in an array and sort the array |
| C. them in a linked list and sort the linked list | D. references to them in an array and sort the array |

nielit2016mar-scientistc data-structures array

Answer key

37.3.2 Array: NIELIT 2017 July Scientist B (CS) - Section B: 4



Let A be a square matrix of size $n \times n$. Consider the following program. What is the expected output?

```

C=100
for i=1 to n do
  for j=1 to n do
  {
    Temp=A[i][j]+C
    A[i][j]=A[j][i]
    A[j][i]=Temp-C
  }
  for i=1 to n do

```

```
for j=1 to n do  
  Output(A[i][j]);
```

- A. The matrix A itself.
- B. Transpose of matrix A .
- C. Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A .
- D. None of the option.

nielit2017july-scientistb-cs data-structures array

Answer key

37.3.3 Array: NIELIT 2017 July Scientist B (CS) - Section B: 40



Kadane algorithm is used to find

- A. Maximum sum subsequence in an array
- B. Maximum sum subarray in an array
- C. Maximum product subsequence in an array
- D. Maximum product subarray in an array

nielit2017july-scientistb-cs data-structures array

Answer key

37.3.4 Array: NIELIT Scientific Assistant A 2020 November: 109



A program P reads in 500 integers in the range $[0 \dots 100]$ representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for P to store the frequencies?

- A. An array of 50 numbers
- B. An array of 100 numbers
- C. An array of 500 numbers
- D. A dynamically allocated array of 550 numbers

nielit-sta-2020 data-structures array easy

Answer key

37.4

Binary Heap (5)

37.4.1 Binary Heap: NIELIT 2016 MAR Scientist C - Section C: 49



We have a binary heap on n elements and wish to insert n more elements (not necessarily one after another) into this heap. Total time required for this is

- A. $\Theta(\log n)$
- B. $\Theta(n)$
- C. $\Theta(n \log n)$
- D. $\Theta(n^2)$

nielit2016mar-scientistc data-structures binary-heap

Answer key

37.4.2 Binary Heap: NIELIT 2017 DEC Scientist B - Section B: 31



Consider a complete binary tree where the left and the right sub trees of the root are max-heaps. The lower bound for the number of operations to convert the tree to a heap is:

- A. $\Omega(\log n)$ B. $\Omega(n \log n)$ C. $\Omega(n)$ D. $\Omega(n^2)$

nielit2017dec-scientistb data-structures binary-tree binary-heap

Answer key

37.4.3 Binary Heap: NIELIT 2018-37



The smallest element that can be found in time _____ in a binary max heap.

- A. $O(n \log n)$ B. $O(\log n)$ C. $O(n)$ D. $O(n^2)$

nielit-2018 data-structures binary-heap

Answer key

37.4.4 Binary Heap: NIELIT 2018-84



For the given nodes:

89, 19, 50, 17, 12, 15, 2, 5, 7, 11, 6, 9, 100

minimum _____ number of interchanges are required to convert it into a max-heap.

- A. 3 B. 4 C. 5 D. 6

nielit-2018 data-structures binary-heap

Answer key

37.4.5 Binary Heap: NIELIT 2022 April Scientist B | Section B | Question: 107



Consider a max heap, represented by the array: 40, 30, 20, 10, 15, 16, 17, 8, 4. Now consider that a value 35 is inserted into this heap. After insertion, the new heap is

- A. 40, 30, 20, 10, 15, 16, 17, 8, 4, 35 B. 40, 35, 20, 10, 30, 16, 17, 8, 4, 15
C. 40, 30, 20, 10, 35, 16, 17, 8, 4, 15 D. 40, 35, 20, 10, 15, 16, 17, 8, 4, 30

nielit2022apr-scientistb data-structures binary-heap algorithms

37.5

Binary Search (1)

37.5.1 Binary Search: NIELIT 2016 MAR Scientist C - Section C: 52



Consider the process of inserting an element into a *Max Heap*, where the *Max Heap* is represented by an *array*. Suppose we perform a binary search on the path from the new leaf to the root to find the position for the newly inserted element, the number of *comparisons* performed is

- A. $\Theta(\log_2 n)$ B. $\Theta(n \log_2 \log_2 n)$

C. $\Theta(n)$

D. $\Theta(n \log_2 n)$

nielit2016mar-scientistc data-structures binary-search time-complexity binary-heap

Answer key

37.6

Binary Search Tree (3)

37.6.1 Binary Search Tree: NIELIT 2016 MAR Scientist C - Section C: 50



You are given the postorder traversal, P , of a binary search tree on the n elements $1, 2, \dots, n$. You have to determine the unique binary search tree that has P as its postorder traversal. What is the time complexity of the most efficient algorithm for doing this?

- A. $\Theta(\log n)$
- B. $\Theta(n)$
- C. $\Theta(n \log n)$
- D. None of the above, as the tree cannot be uniquely determined.

nielit2016mar-scientistc data-structures binary-search-tree

Answer key

37.6.2 Binary Search Tree: NIELIT 2018-67



_____ is the worst-case time complexity for all operations (i.e., search, update and delete) in a general Binary Search tree

- A. $O(n)$
- B. $O(n \log n)$
- C. $O(\log n)$ for search and insert, and $O(n)$ for delete
- D. None of these

nielit-2018 data-structures binary-search-tree

Answer key

37.6.3 Binary Search Tree: NIELIT Scientific Assistant A 2020 November: 93



When we perform in order traversal on a binary tree, we get the ascending order array. The tree is:

- A. Heap tree
- B. almost complete binary tree
- C. Binary search tree
- D. Cannot be determined

nielit-sta-2020 data-structures binary-search-tree

Answer key

37.7

Binary Tree (11)

37.7.1 Binary Tree: NIELIT 2016 DEC Scientist B (CS) - Section B: 15



Which of the following need not be a binary tree?

- A. Search tree
- B. Heap
- C. AVL tree
- D. B tree

Answer key **37.7.2 Binary Tree: NIELIT 2016 DEC Scientist B (IT) - Section B: 13**

The number of unused pointers in a complete binary tree of depth 5 is:

- A. 4 B. 8 C. 16 D. 32

Answer key **37.7.3 Binary Tree: NIELIT 2016 MAR Scientist C - Section C: 31**

A full binary tree with n non-leaf nodes contains

- A. $\log_2 n$ nodes B. $n + 1$ nodes
C. $2n$ nodes D. $2n + 1$ nodes

Answer key **37.7.4 Binary Tree: NIELIT 2017 DEC Scientific Assistant A - Section B: 18**

In a full binary tree number of nodes is 63 then the height of the tree is :

- A. 2 B. 4 C. 3 D. 6

Answer key **37.7.5 Binary Tree: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 2**

The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number of nodes in a binary tree of height h is

- A. 2^h B. $2^{h-1} - 1$ C. $2^{h+1} - 1$ D. 2^{h+1}

Answer key **37.7.6 Binary Tree: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 20**

The number of possible binary trees with 4 nodes is

- A. 12 B. 13 C. 14 D. 15

Answer key 

37.7.7 Binary Tree: NIELIT 2021 Dec Scientist A - Section B: 76



When the left sub-tree of the tree is one level higher than that of the right sub-tree, then the balance factor is _____.

- A. 0 B. 1 C. -1 D. 2

nielit2021dec-scientista data-structures binary-tree

Answer key

37.7.8 Binary Tree: NIELIT 2021 Dec Scientist A - Section B: 78



Total number of nodes at the n^{th} level of a full binary tree can be given as _____.

- A. $2n+1$ B. $2n^2$ C. 2^n D. $2n-1$

nielit2021dec-scientista binary-tree data-structures

Answer key

37.7.9 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 38



The maximum number of binary trees that can be formed with three unlabeled nodes is :

- A. 1 B. 5 C. 4 D. 3

nielit-2021-it-dec-scientistb binary-tree combinatory data-structures algorithms

Answer key

37.7.10 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 51



In a binary tree, the number of internal nodes of degree 1 is 5, and the number of nodes of degree 2 is 10. The number of leaf nodes in binary tree is:

- A. 10 B. 11 C. 12 D. 15

nielit-2021-it-dec-scientistb data-structures binary-tree

Answer key

37.7.11 Binary Tree: NIELIT 2021 Dec Scientist B - Section B: 90



A binary tree of depth K is called a full binary tree of depth K , if it has exactly _____ nodes.

- A. K B. 2^k C. 2^k-1 D. 2^k+1

nielit2021dec-scientistb binary-tree data-structures

37.8.1 Data Structures: NIELIT 2018-70



_____ number of underlined graphs can be constructed using $V = (v_1, v_2, \dots, v_n)$.

- A. n^3 B. $2^{n(n-1)}/2$ C. $n - 1/2$ D. $2^{(n-1)}/2$

nielit-2018 data-structures graph-theory

Answer key

37.9 Depth First Search (3)

37.9.1 Depth First Search: NIELIT 2016 DEC Scientist B (IT) - Section B: 30



What data structures is used for depth first traversal of a graph?

- A. Queue B. Stack C. List D. None of the above

nielit2016dec-scientistb-it data-structures stack depth-first-search

Answer key

37.9.2 Depth First Search: NIELIT 2016 DEC Scientist B (IT) - Section B: 42



In the _____ traversal we process all of a vertex's descendants before we move to an adjacent vertex.

- A. Depth First B. Breadth First C. Width First D. Depth Limited

nielit2016dec-scientistb-it data-structures stack depth-first-search

Answer key

37.9.3 Depth First Search: NIELIT 2017 July Scientist B (IT) - Section B: 4



What are the appropriate data structures for graph traversal using Breadth First Search(BFS) and Depth First Search(DFS) algorithms?

- A. Stack for BFS and Queue for DFS B. Queue for BFS and Stack for DFS
C. Stack for BFS and Stack for DFS D. Queue for BFS and Queue for DFS

nielit2017july-scientistb-it data-structures graph-algorithms breadth-first-search depth-first-search

Answer key

37.10 Expression Evaluation (2)

37.10.1 Expression Evaluation: NIELIT 2018-54



_____ to evaluate an expression without any embedded function calls.

- A. Two stacks are required B. one stack is needed
C. Three stacks are required D. More than three stacks are required

nielit-2018 data-structures stack expression-evaluation

Answer key 

37.10.2 Expression Evaluation: NIELIT 2018-85



Evaluation of the given postfix expression $10\ 10 + 60\ 6 / * 8 -$ is

- A. 192 B. 190 C. 110 D. 92

nielit-2018 data-structures expression-evaluation

Answer key 

37.11

Hashing (5)

37.11.1 Hashing: NIELIT 2016 MAR Scientist C - Section C: 30



A hash table with 10 buckets with one slot per bucket is depicted. The symbols, $S1$ to $S7$ are initially emerged using a hashing function with linear probing. Maximum number of comparisons needed in searching an item that is not present is

0	$S7$
1	$S1$
2	
3	$S4$
4	$S2$
5	
6	$S5$
7	
8	$S6$
9	$S3$

- A. 6 B. 5 C. 4 D. 3

nielit2016mar-scientistc data-structures hashing

Answer key 

37.11.2 Hashing: NIELIT 2016 MAR Scientist C - Section C: 56



The average search time of hashing, with linear probing will be less if the load factor

- A. is far less than one B. equals one
C. is far greater than one D. none of these

nielit2016mar-scientistc data-structures hashing

Answer key 

37.11.3 Hashing: NIELIT 2018-65



For a given hash table T with 10 slots that stores 1000 elements, the load factor α for T is

- A. 100 B. 0.01 C. 200 D. 1.05

nielit-2018 data-structures hashing

Answer key 

37.11.4 Hashing: NIELIT 2021 Dec Scientist A - Section B: 51



Which open addressing technique is free from Clustering problems?

- A. Linear probing
- B. Quadratic probing
- C. Double hashing
- D. Rehashing

nielit2021dec-scientista hashing data-structures algorithms

Answer key 

37.11.5 Hashing: NIELIT Scientist B 2020 November: 73



Which open addressing technique is free from Clustering problems?

- A. Linear probing
- B. Quadratic probing
- C. Double hashing
- D. Rehashing

nielit-scb-2020 hashing data-structures

Answer key 

37.12

Infix Prefix (1)

37.12.1 Infix Prefix: NIELIT 2017 July Scientist B (CS) - Section B: 6



Assume that the operators $+$, $-$, $\times$ are left associative and $\wedge$ is right associative. The order of precedence (from highest to lowest) is $\wedge$, $\times$, $+$, $-$. The postfix expression corresponding to the infix expression $a + b \times c - d \wedge e \wedge f$ is

- A. $abc \times + def \wedge \wedge -$
- B. $abc \times + de \wedge f \wedge -$
- C. $ab + c \times d - e \wedge f \wedge$
- D. $- + a \times bc \wedge \wedge def$

nielit2017july-scientistb-cs data-structures stack infix-prefix

Answer key 

37.13

Linked List (7)

37.13.1 Linked List: NIELIT 2016 MAR Scientist C - Section C: 54



In a circularly linked list organization, insertion of a record involves the modification of

- A. no pointer
- B. 1 pointer
- C. 2 pointers
- D. 3 pointers

nielit2016mar-scientistc data-structures linked-list

Answer key 

37.13.2 Linked List: NIELIT 2017 DEC Scientific Assistant A - Section B: 52



The address field of linked list :

- A. Contain address of next node
- B. May contain null character
- C. Contain address of next pointer
- D. Both (A) and (B)

Answer key

37.13.3 Linked List: NIELIT 2017 July Scientist B (CS) - Section B: 1



What does the following function do for a given Linked List with first node as head?

```
void fun1(struct node* head)
{
    if(head==NULL)
        return;
    fun1(head->next);
    printf("%d",head->data);
}
```

- A. Prints all nodes of linked lists
- B. Prints all nodes of linked list in reverse order
- C. Prints alternate nodes of Linked List
- D. Prints alternate nodes in reverse order

37.13.4 Linked List: NIELIT 2017 July Scientist B (CS) - Section B: 3



Consider the following function that takes reference to head of a Doubly Linked List as parameter. Assume that a node of doubly linked list has previous pointer as *prev* and next pointer as *next*.

```
void fun(struct node **head_ref)
{
    struct node *temp=NULL;
    struct node *current=*head_ref;
    while(current!=NULL)
    {
        temp=current->prev;
        current->prev=current->next;
        current->next=temp;
        current=current->prev;
    }
    if(temp!=NULL)
        *head_ref=temp->prev;
}
```

Assume that reference of head of following doubly linked list is passed to above function
 $1 \leftrightarrow 2 \leftrightarrow 3 \leftrightarrow 4 \leftrightarrow 5 \leftrightarrow 6$. What should be the modified linked list after the function call?

- A. $2 \leftrightarrow 1 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 6 \leftrightarrow 5$
- B. $5 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 2 \leftrightarrow 1 \leftrightarrow 6$
- C. $6 \leftrightarrow 5 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 2 \leftrightarrow 1$
- D. $6 \leftrightarrow 5 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 1 \leftrightarrow 2$

37.13.5 Linked List: NIELIT 2021 Dec Scientist A - Section B: 104



Which type of linked list stores the address of the header node in the next field of the last node ?

- A. Singly linked list
- B. Circular linked list
- C. Doubly linked list
- D. Circular header linked list

nielit2021dec-scientista data-structures linked-list

Answer key

37.13.6 Linked List: NIELIT 2021 Dec Scientist B - Section B: 48



The following type definition is for a _____ .

type pointer = ↑ node

node = record

data : integer

link : pointer

end;

- A. Structure
- B. Link List
- C. Stack
- D. Doubly link list

nielit2021dec-scientistb data-structures linked-list programming-in-c

Answer key

37.13.7 Linked List: NIELIT Scientist B 2020 November: 90



Which type of linked list stores the address of the header node in the next field of the last node?

- A. Singly linked list
- B. Circular linked list
- C. Doubly linked list
- D. Hashed list

nielit-scb-2020 linked-list data-structures

Answer key

37.14 Operator Precedence (2)

37.14.1 Operator Precedence: NIELIT 2016 DEC Scientist B (IT) - Section B: 17



Which of the following is the correct order of evaluation for the below expression?

$$z = x + y * z / 4 \% 2 - 1$$

- A. $*/\% + - =$
- B. $=* /\% + -$
- C. $/*\% - + =$
- D. $*/\% / - + =$

nielit2016dec-scientistb-it data-structures operator-precedence stack

Answer key

37.14.2 Operator Precedence: NIELIT 2017 DEC Scientific Assistant A - Section B: 59

The expression $5 - 2 - 3 * 5 - 2$ will evaluate to 18, if :



- A. '-' is left associative and '*' has precedence over '-'
- B. '-' is right associative and '*' has precedence over '-'
- C. '-' is right associative and '-' has precedence over '*'
- D. '-' is left associative and '-' has precedence over '*'

nielit2017dec-assistanta data-structures operator-precedence

Answer key

37.15

Pre Order Traversal (1)

37.15.1 Pre Order Traversal: NIELIT Scientist B 2020 November: 63



Binary search tree contains the values 1, 2, 3, 4, 5, 6, 7, 8. The tree is traversed in pre-order and the values are printed out. Which of the following sequences is a valid output?

- A. 53124786
- B. 53126487
- C. 53241678
- D. 53124768

nielit-scb-2020 binary-search-tree tree data-structures pre-order-traversal

Answer key

37.16

Priority Queue (1)

37.16.1 Priority Queue: NIELIT 2017 July Scientist B (CS) - Section B: 8



A priority queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is: 10, 8, 5, 3, 2. Two new elements 1 and 7 are inserted into the heap in that order. The level-order traversal of the heap after the insertion of the elements is

- A. 10, 8, 7, 3, 2, 1, 5
- B. 10, 8, 7, 2, 3, 1, 5
- C. 10, 8, 7, 1, 2, 3, 5
- D. 10, 8, 7, 5, 3, 2, 1

nielit2017july-scientistb-cs data-structures priority-queue binary-heap

Answer key

37.17

Queue (10)

37.17.1 Queue: NIELIT 2016 DEC Scientist B (IT) - Section B: 15



A _____ is a linear list in which insertions and deletions are made to from either end of the structure.

- A. Circular queue.
- B. Priority queue.
- C. Stack.
- D. Dequeue.

nielit2016dec-scientistb-it data-structures queue

Answer key

37.17.2 Queue: NIELIT 2017 July Scientist B (CS) - Section B: 10



A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT(n refers to the number of items in the queue)?

- A. Both operations can be performed in $O(1)$ time.
- B. At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$.
- C. The worst case time complexity for both operations will be $\Omega(n)$.
- D. Worst case time complexity for both operations will be $\Omega(\log n)$.

nielit2017july-scientistb-cs data-structures queue

Answer key 

37.17.3 Queue: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 12



If queue is implemented using arrays, what would be the worst run time complexity of queue and dequeue operations?

- A. $O(n), O(n)$ B. $O(n), O(1)$ C. $O(1), O(n)$ D. $O(1), O(1)$

nielit2017oct-assistanta-it data-structures queue

Answer key 

37.17.4 Queue: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 6



Which of the following is useful in traversing a given graph by breadth first search?

- A. Stack B. Set C. List D. Queue

nielit2017oct-assistanta-it data-structures queue

Answer key 

37.17.5 Queue: NIELIT 2018-69



_____ number of queues are needed to implement a stack

- A. 1 B. 2 C. 3 D. 4

nielit-2018 data-structures queue

Answer key 

37.17.6 Queue: NIELIT 2021 Dec Scientist B - Section B: 19



In which types of following data structure, an element is inserted at one end called Rear and deleted at other end called Front?

- A. Stack B. Queue
C. Height balanced Avl tree D. Binary tree

37.18.1 Recurrence Relation: NIELIT 2016 DEC Scientist B (IT) - Section B: 16



The recurrence relation capturing the optimal execution time of the Towers of Hanoi problem with n discs is:

A. $T(n) = 2T(n-2) + 2$
C. $T(n) = 2T(n-1) + n$

B. $T(n) = 2T(n/2) + 1$
D. $T(n) = 2T(n-1) + 1$

nielit2016dec-scientistb-it data-structures recurrence-relation

Answer key

37.19

Searching (1)

37.19.1 Searching: NIELIT 2021 Dec Scientist A - Section B: 55



If a hash table is implemented as a search tree, the expected time required to enter n names and make m searches is proportional to :

A. $(n+m)\log_2 n$
C. $mn\log_2 n$

B. $(n+m)\log_2 m$
D. $mn\log_2 m$

nielit2021dec-scientista data-structures searching time-complexity algorithm-design

Answer key

37.20

Sorting (1)

37.20.1 Sorting: NIELIT Scientist B 2020 November: 82



Which of the following techniques deals with sorting the data stored in the computer's memory?

A. Distribution sort
C. External sort

B. Internal sort
D. Radix sort

nielit-scb-2020 sorting data-structures

Answer key

37.21

Stack (2)

37.21.1 Stack: NIELIT 2022 April Scientist B | Section B | Question: 106



A single array $A[1 \dots \text{MAXSIZE}]$ is used to implement two stacks. The two stacks grow from opposite ends of the array. Variables top 1 and top 2 ($\text{top 1} < \text{top 2}$) point to the location of the topmost element in each of the stacks. If the space is to be used efficiently, the condition for "stack full" is

A. $(\text{top 1} = \text{MAXSIZE}/2)$
 $(\text{top 2} = \text{MAXSIZE}/2 + 1)$

or $\text{top 1} + \text{top 2} = \text{MAXSIZE}$

C. $(\text{top 1} = \text{MAXSIZE}/2)$
 $(\text{top 2} = \text{MAXSIZE})$

or $\text{top 1} = \text{top 2} - 1$

nielit2022apr-scientistb data-structures stack array

37.21.2 Stack: NIELIT Scientific Assistant A 2020 November: 115



A recursive problem like tower of hanoi can be rewritten without recursion using:

- A. stack B. priority queue C. graph D. cycles

nielit-sta-2020 data-structures stack

Answer key

37.22

Tree (6)

37.22.1 Tree: NIELIT 2016 DEC Scientist B (CS) - Section B: 42



The maximum number of nodes in a binary tree of level k , $k \geq 1$ is:

- A. $2^k + 1$ B. 2^{k-1} C. $2^k - 1$ D. $2^{k-1} - 1$

nielit2016dec-scientistb-cs data-structures tree binary-tree

Answer key

37.22.2 Tree: NIELIT 2017 July Scientist B (CS) - Section B: 54



In a complete k -ary tree, every internal node has exactly k children. The number of leaves in such a tree with n internal nodes is

- A. nk B. $(n - 1)k + 1$ C. $n(k - 1) + 1$ D. $n(k - 1)$

nielit2017july-scientistb-cs data-structures tree

Answer key

37.22.3 Tree: NIELIT 2018-72



_____ number of leaf nodes in a rooted tree of n nodes, where each node is having 0 or 3 children

- A. $\frac{n}{2}$ B. $\frac{(2n+1)}{3}$ C. $\frac{(n-1)}{n}$ D. $(n - 1)$

nielit-2018 data-structures tree

Answer key

37.22.4 Tree: NIELIT 2021 Dec Scientist A - Section B: 47



Suppose a binary search tree has been constructed from the following sequence of numbers in the order in which they arrive : 6, 2, 10, 1, 5, 7, 11, 3, 9, 4, 8. Consider the following piece of code :

```
Show(root) { if (root != NULL)
    {printf("%d", root → key);
    show (root → right);
    show (root → left);
    }
```



```
        else
    return ;
}
```

The sequence printed will be :

- A. 6,11,10,7,8,9,2,4,3,5,1 B. 6,11,7,9,8,10,2,5,1,3,4
C. 6,10,11,7,9,8,2,5,3,4,1 D. 6,10,2,11,7,9,8,5,3,4,1

nielit2021dec-scientista binary-search-tree tree binary-tree data-structures programming-in-c

Answer key 

37.22.5 Tree: NIELIT 2022 April Scientist B | Section B | Question: 59



Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are:

The height of a tree with a single node is 0.

- A. 4 and 15 respectively B. 3 and 14 respectively
C. 4 and 14 respectively D. 3 and 15 respectively

nielit2022apr-scientistb binary-search-tree data-structures tree

Answer key 

37.22.6 Tree: NIELIT Scientist B 2020 November: 106



Suppose we have to insert the following sequence of keys into an empty binary search tree:

5, 7, 45, 60, 50, 23, 15, 54

What would be the height of binary search tree?

- A. 3 B. 4 C. 5 D. 6

nielit-scb-2020 binary-search-tree data-structures tree

Answer key 

37.23 Tree Traversal (7)

37.23.1 Tree Traversal: NIELIT 2016 MAR Scientist B - Section C: 18



Traversing a binary tree first root and then left and right subtrees called _____ traversal.

- A. postorder. B. preorder. C. inorder. D. none of these.

nielit2016mar-scientistb data-structures binary-tree tree-traversal

Answer key 

37.23.2 Tree Traversal: NIELIT 2017 DEC Scientific Assistant A - Section B: 2



In binary search tree which traversal is used for getting ascending order values ?

- A. Inorder B. Preorder C. Postorder D. None of the options

nielit2017dec-assistanta data-structures binary-search-tree tree-traversal

Answer key 

37.23.3 Tree Traversal: NIELIT 2017 DEC Scientist B - Section B: 30



If for a given Binary Search Tree (BST) the pre-order traversal is 41, 23, 11, 31, 62, 50, 73. Then which of the following is its post-order traversal?

- A. 11, 31, 23, 50, 73, 62, 41 B. 31, 11, 23, 50, 41, 62, 73
C. 11, 31, 50, 23, 73, 62, 41 D. 11, 31, 23, 50, 62, 73, 41

nielit2017dec-scientistb data-structures binary-search-tree tree-traversal

Answer key 

37.23.4 Tree Traversal: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 5



A binary search tree contains the values-1, 2, 3, 4, 5, 6, 7 and 8. The tree is traversed in preorder and the values are printed out. Which of the following sequences is a valid output?

- A. 5 3 1 2 4 7 8 6 B. 5 3 1 2 6 4 9 7
C. 5 3 2 4 1 6 7 8 D. 5 3 1 2 4 7 6 8

nielit2017oct-assistanta-cs data-structures binary-search-tree tree-traversal

Answer key 

37.23.5 Tree Traversal: NIELIT 2018-80



_____ traversals are not sufficient to build a binary tree.

- A. Preorder and Inorder B. Postorder and Inorder
C. Postorder and Preorder D. None of these

nielit-2018 data-structures tree-traversal

Answer key 

37.23.6 Tree Traversal: NIELIT 2021 Dec Scientist B - Section B: 74



The preorder traversal of a binary search tree is given by 12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20.

Then postorder traversal will be:

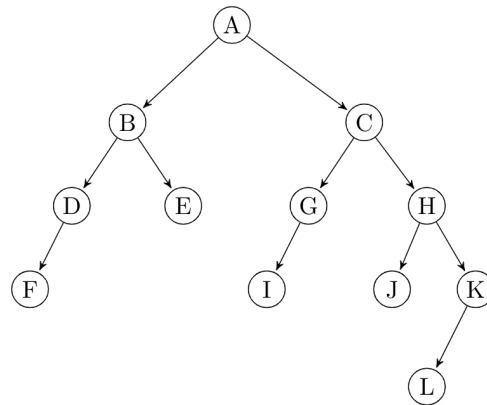
- A. 2, 6, 7, 8, 9, 10, 12, 15, 16, 17, 19, 20
B. 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12
C. 7, 2, 6, 8, 9, 10, 20, 17, 19, 15, 16, 12
D. 7, 6, 2, 10, 9, 8, 15, 16, 17, 20, 19, 12

nielit-2021-it-dec-scientistb binary-search-tree binary-tree tree-traversal data-structures algorithm-design

Answer key 



The Preorder traversal of a tree given below is:



A. ABDFECGIHJKL
C. ABEDFCGHIJKL

B. ABCDEGHIJFKL
D. ABDFECGIJHKL

nielit-scb-2020 tree-traversal tree data-structures

Answer key

Answer Keys

37.1.1	D	37.2.1	B	37.3.1	B	37.3.2	A	37.3.3	B
37.3.4	A	37.4.1	B	37.4.2	A	37.4.3	C	37.4.4	A
37.4.5	B	37.5.1	B	37.6.1	B	37.6.2	A	37.6.3	C
37.7.1	D	37.7.2	D	37.7.3	D	37.7.4	X	37.7.5	C
37.7.6	C	37.7.7	B	37.7.8	C	37.7.9	B	37.7.10	B
37.7.11	C	37.8.1	B	37.9.1	B	37.9.2	A	37.9.3	B
37.10.1	B	37.10.2	A	37.11.1	X	37.11.2	A	37.11.3	A
37.11.4	C	37.11.5	C	37.12.1	A	37.13.1	C	37.13.2	D
37.13.3	B	37.13.4	C	37.13.5	B	37.13.6	B	37.13.7	B
37.14.1	A	37.14.2	A	37.15.1	D	37.16.1	A	37.17.1	D
37.17.2	A	37.17.3	D	37.17.4	D	37.17.5	B	37.17.6	B
37.17.7	B	37.17.8	C	37.17.9	B	37.17.10	A	37.18.1	D
37.19.1	A	37.20.1	B	37.21.1	D	37.21.2	A	37.22.1	C
37.22.2	C	37.22.3	B	37.22.4	C	37.22.5	B	37.22.6	D
37.23.1	B	37.23.2	A	37.23.3	A	37.23.4	D	37.23.5	C
37.23.6	B	37.23.7	A						